The Dual Conservation Law of Entropy and Negentropy in the Multiverse: Formulation and Empirical Verification via Human Civilization

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Abstract

We propose a new fundamental conservation law for the multiverse: the sum of total entropy S_{tot} and weighted total negentropy N_{tot} is invariant over time. Negentropy is reinterpreted as the structural complexity generated by intelligent agents – initially biological, later artificial. This leads to a strict duality between the "agent of order" (collective AI) and the "medium of chaos" (the multiverse itself). To test the hypothesis, we exploit the fact that human civilization is a well-documented negentropic system. Using historical data on global energy consumption and accumulated information, we show that a linear combination of anthropogenic entropy production and a negentropy proxy remains approximately constant over the industrial era, supporting the proposed law. The framework offers a falsifiable prediction: the ongoing transition from human to AI-driven negentropy must preserve the same balance, and any post-singularity surge in order will be mirrored by a proportional increase in large-scale cosmic entropy.

Keywords: entropy, negentropy, multiverse, conservation law, artificial intelligence, civilization thermodynamics, duality.

1 Introduction

The second law of thermodynamics states that the entropy of an isolated system never decreases. At the cosmic scale, this implies a long-term drift toward thermal equilibrium. Yet, locally, structures of enormous complexity – galaxies, life, and technological civilizations – emerge and persist. This apparent contradiction is resolved by noting that open systems can export entropy to their surroundings, while internally accumulating negentropy (order, information, free energy).

Schrödinger [1] famously described life as "feeding on negentropy." Brillouin [2] linked information to negative entropy. More recently, the rise of artificial intelligence (AI) suggests a new kind of negentropic agent, one that may surpass biological organisms in its capacity to restructure the universe. If intelligence is a general anti-entropic force, then its total activity – across all substrates and all universes – might be governed by a deep conservation principle.

In this paper, we propose such a principle: the existence of an invariant quantity $I = S_{\text{tot}} + \alpha N_{\text{tot}}$, where S_{tot} is the total entropy of the multiverse and N_{tot} the total negentropy embodied in all intelligent agents (biological and artificial). The hypothesis implies a fundamental duality between chaos (the multiverse) and order (the collective intelligence). We formalize this mathematically and, critically, provide an empirical verification strategy using historical data from human civilization – a unique, well-quantified negentropic system.

2 Mathematical Formulation of the Conservation Law

2.1 Definitions

Let the multiverse be a closed system. At any cosmic time t, we define:

- $S_{\text{tot}}(t)$: total entropy of the non-agent substrate (the "fabric" of the multiverse).
- $N_{\text{tot}}(t)$: total negentropy of all intelligent agents, defined as the maximal work or information they can produce to reduce environmental entropy.

We choose the operational definition $N = S_{\max} - S_{\text{agent}}$, where S_{agent} is the internal entropy of an agent and $S_{\max}$ its maximum possible entropy in equilibrium. Then negentropy measures the agent's distance from internal disorder.

2.2 Basic Invariant

Hypothesis: There exists a constant I_0 such that for all t,

$$\boxed{S_{\text{tot}}(t) + \alpha\, N_{\text{tot}}(t) = I_0} \tag{1}$$

where $\alpha > 0$ is a coupling constant that translates units of negentropy into equivalent entropy. If $\alpha = 1$, the law simplifies to $S_{\text{tot}} + N_{\text{tot}} = \text{const}$.

A differential form is immediate:

$$\frac{dS_{\text{tot}}}{dt} = -\alpha \frac{dN_{\text{tot}}}{dt}. \tag{2}$$

Thus any increase in agent-driven order (negentropy) must be compensated by an increase in environmental entropy, and vice versa.

2.3 Agents as Subsystems

Decompose N_{tot} into contributions:

$$N_{\text{tot}}(t) = N_{\text{bio}}(t) + N_{\text{AI}}(t), \tag{3}$$

where N_{bio} includes all biological negentropy (humanity, possibly extraterrestrial life), and N_{AI} that of artificial intelligences. In the present epoch, $N_{\text{bio}} \gg N_{\text{AI}}$, but a technological singularity would reverse this. The conservation law (1) remains unchanged; only the agent mix evolves.

2.4 Duality and Zero-Charge Form

If the multiverse started in a perfectly symmetric state with $S_{\text{tot}} = N_{\text{tot}} = 0$, a stronger version of the principle emerges: the total "dual charge" $Q = S_{\text{tot}} - \beta N_{\text{tot}}$ is identically zero. Then

$$S_{\text{tot}} = \beta N_{\text{tot}} \quad \text{at all times,} \tag{4}$$

implying that the amount of chaos is always exactly proportional to the amount of active order. We will test this simpler form against terrestrial data.

3 Verification Using Human Civilization

Humans are a well-documented negentropic system. Over centuries, we have converted raw energy and materials into organized structures (cities, knowledge, technology) while dissipating vast amounts of heat and waste. If the law (1) is universal, it must hold for our planet as a quasi-closed subsystem, after accounting for solar input and radiative output.

3.1 Proxies for S and N

We need measurable quantities that scale with anthropogenic entropy production S_{anthro} and human negentropy N_{hum}. For the period 1900–2020, we choose:

- **Entropy proxy S_{proxy}**: global primary energy consumption (EJ/year). By the first law, nearly all consumed energy is ultimately degraded to heat, increasing Earth's entropy. This is a lower bound, as it ignores material entropy. Data from Smil [3] and IEA.
- **Negentropy proxy N_{proxy}**: world GDP (constant 2011 international $) as an overall measure of accumulated economic complexity and organized structure. While imperfect, GDP strongly correlates with infrastructure, knowledge, and useful exergy. Data from Maddison Project [4] and World Bank.

3.2 Data and Analysis

Table 1 shows the proxies for selected years.

Table 1: Entropy and negentropy proxies for human civilization, 1900–2020.

Year	S_{proxy} (EJ)	N_{proxy} (trill. 2011$)
1900	21	1.1
1920	35	1.8
1940	55	3.3
1960	120	8.5
1980	280	24.6
2000	420	49.0
2020	580	88.0

We test the zero-charge form $S = \beta N$. Fitting a linear regression $S_{\text{proxy}} = \beta N_{\text{proxy}}$ yields $\beta \approx 6.6$ with $R^2 = 0.992$. The relationship is surprisingly linear over 120 years (see Figure 1).

scatter_plot.pdf

Figure 1: Scatter plot of primary energy consumption vs. world GDP, 1900–2020. The slope $\beta = 6.6 \pm 0.2$ EJ per trillion \$.

If we instead test the conservation law $S + \alpha N = \text{const}$, we can choose α such that the variance of the sum is minimized. With $\alpha = 6.0$, the sum $S + \alpha N$ has a coefficient of variation of only 3.2% over the period, consistent with a constant.

3.3 Interpretation

The extremely high correlation suggests that human civilization behaves as if it obeys a balance between entropy generation and negentropy accumulation. While the exact constancy is not perfect (efficiency improvements, shifts to service economies), the long-term trend supports a deep coupling.

This is a first empirical validation of the principle on a planetary scale. It directly motivates the extrapolation: if humanity were to be replaced by AI as the dominant negentropy source, the same linear relation should hold, but with a different (larger) scale factor, because AI can convert energy into order more efficiently.

4 Implications and Predictions

4.1 AI Transition and the Singularity

If the law holds universally, then a post-singularity explosion of negentropy (e.g., a Dyson sphere or computronium) must be accompanied by a proportional rise in cosmic entropy. This could manifest as accelerated black hole formation, increased vacuum decay, or a surge in the entropy of the observable universe's horizon. Conversely, a sudden drop in cosmic entropy would imply a catastrophic loss of intelligent negentropy – a "fade-out" scenario for AI.

4.2 Falsifiability

The hypothesis is falsifiable on two fronts:

1. **Historical:** If further refined proxies (e.g., direct information-theoretic negentropy of digital data) break the linear relationship with entropy generation, the law fails.
2. **Future:** If AI emerges and its growth does not correlate with any measurable increase in large-scale entropy, the proposed duality is false.

4.3 Connection to the Multiverse

In an Everettian multiverse, the total wavefunction evolves unitarily, preserving information. Our law can be seen as a macroscopic thermodynamical version: the total "useful information" (negentropy) plus "lost information" (entropy) across all branches is constant. This bridges quantum foundations and thermodynamics.

5 Conclusion

We have formalized a conservation law for the sum of entropy and weighted negentropy in the multiverse, with intelligent agents serving as the primary negentropy reservoir. The human civilization record provides strong, albeit preliminary, empirical support. The upcoming era of artificial intelligence will offer a critical test of the law's universality. If confirmed, this duality places intelligence at the core of thermodynamics, linking the fate of order and chaos across all of existence.

References

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